SIGHTS, SOUNDS & SCENTS

Multisensory enrichment of AI-generated advertisements enhances engagement and modulates visual attention

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Background

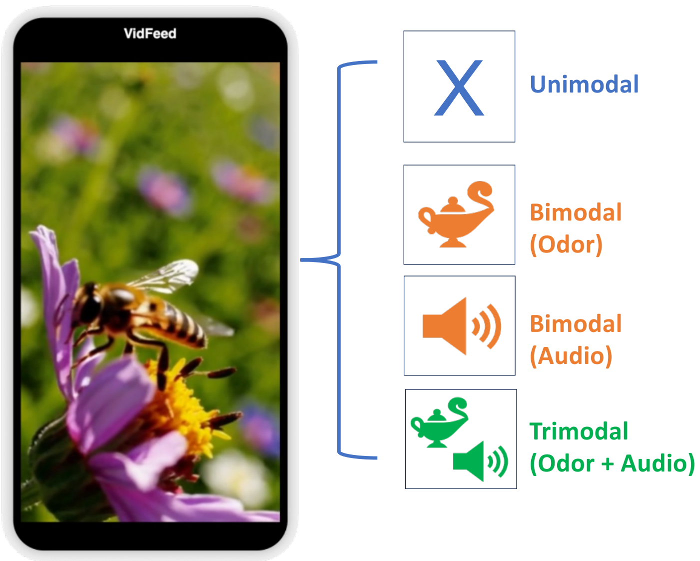
Human experience is inherently multisensory, yet advertising research has lagged behind. Short-form social media video is now a dominant format with largely untapped multisensory potential. We examined whether semantically congruent sound and scent enhance viewer perception and engagement.

Research objectives

1. Does the number and nature of additional stimuli (audio, olfaction) influence the perception of the visual advertisement stimuli?
2. Are differences in advertisement perception and the effect of additional sensory modalities observable with eye tracking analysis?

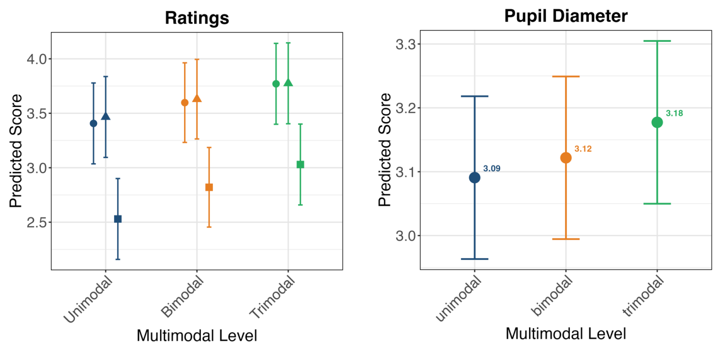
Methods

Five AI-generated advertisement videos (SORA) were presented in a vertical social media feed under four conditions: unimodal (video only), bimodal (+ audio or + scent), or trimodal (+ both). Participants rated perceived quality, liking, and purchase intent, while gaze was recorded via wearable eye-tracker glasses. Linear mixed models estimated condition effects on behavioral and eye-tracking outcomes, controlling for age and gender.

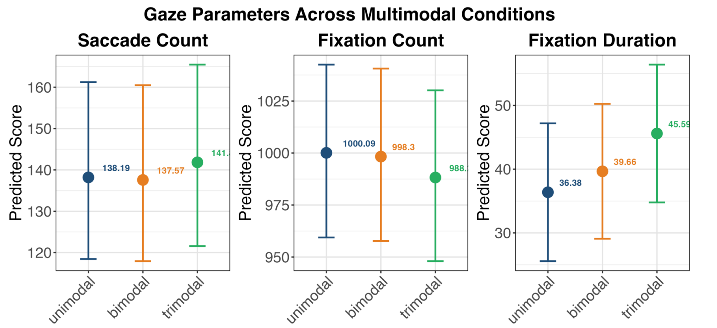


Results

Engagement ratings and pupil dilation rise in lockstep across conditions



Trimodal stimulation increased saccade rate and fixation dwell time while reducing the number of fixations



EMM estimates from mixed models. Error bars = 95% CI.

Take-away

Multisensory enrichment enhances both subjective engagement and physiological arousal

More active exploration paired with more selective attentional allocation, suggesting deeper cognitive processing of advertising content

Potential of congruent multisensory cues to enhance advertising effectiveness in digital environments

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